

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Topic 3.1 - Air pollution

A real classroom situation

Every winter, schools in Lahore watch the sky turn the colour of dishwater. In November, Miss Farzana's cousin, a headteacher there, checks the air quality index each morning the way coastal people check the tide. On a bad day the AQI passes 300, "hazardous", and the government closes the schools. Her pupils stay home, and the news argues about whose fault the smog is: India's crop burning, the brick kilns, the traffic, the weather.

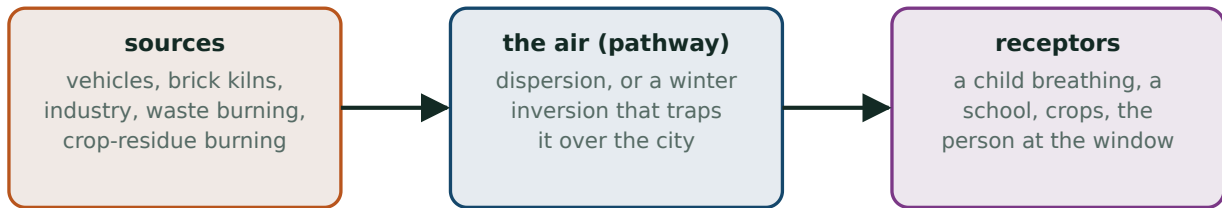
This topic gives you the tools to read that argument properly: what is in the air, where it comes from, what it does to people, and what can actually be done about it.

Explanation

Air pollution: causes, effects and control strategies

Air pollution is the presence of substances in the air at levels that harm people, other living things or materials. Its logic is the same chain you met for water in Unit 2, with air as the pathway.

The air pollution chain, and where controls act



Control at the source (strongest)

cleaner fuel; upgraded kilns; engine standards; an end to open burning

Control in the pathway (weak)

dispersion only moves the pollutant; in an inversion there is little to help

Control at the receptor (a teacher's tools)

AQI check; morning timing of sports; windows shut on hazardous days; asthma plans

The honest answer: the pathway cannot be cleaned, only shortened by cutting emissions.

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The pollutants that matter. Particulate matter, PM10 and the finer PM2.5, is soot, dust and droplets small enough to reach deep into the lungs; it is the pollutant most linked to the health burden. Nitrogen dioxide and sulfur dioxide come from burning fuel at high temperature, in engines and furnaces. Carbon monoxide is the silent one: colourless, odourless, and it takes up the seat in blood that oxygen needs. Ground-level ozone is not emitted directly; sunlight cooks it out of other pollutants, which is why it peaks in hot afternoons. Lead, from some fuels and paints, damages the developing brain; there is no safe level.

The six pollutants that matter

Pollutant	Main source	Health effect	One control
PM2.5 and PM10	combustion, dust, burning	deep-lung and heart disease	end open burning
Nitrogen dioxide	engines, furnaces	airway inflammation, asthma	engine standards
Sulfur dioxide	coal, oil, kilns	airway irritation, acid rain	cleaner fuel, kiln upgrade
Carbon monoxide	incomplete burning, generators	takes oxygen's seat in blood	ventilation, never indoors
Ground-level ozone	sunlight acting on other gases	lung irritation, crop damage	get the gases it cooks from
Lead	some fuels, old paint	damages the developing brain	unleaded fuel

PM2.5 is invisible particle by particle: a clear sky can still carry an unhealthy index.

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How big is the problem? The World Health Organization estimates ambient, outdoor, air pollution caused 4.2 million premature deaths worldwide in 2019, and that 99 per cent of the world's people breathe air exceeding its guideline levels (WHO, 2024). Its 2021 global air quality guidelines set the levels the table's controls are trying to reach (WHO, 2021).

The worked case: Lahore's winter smog. A 2026 open-access study combined satellite observations over winter seasons (2019 to 2021) with a survey of the people who work outdoors in smog season, labourers, street vendors and traffic police, and found winter smog critical for both environment and public health in Lahore district (Anwar et al., 2026). A temperature inversion, cold air trapped under warm, holds the city's own emissions over it. The causes are shared: vehicle exhaust, brick kilns, industry, waste burning and crop-residue burning across the region. No single owner, no single fix.

Control strategies sit at the three links. At the source: cleaner fuel, upgraded kilns, engine standards, an end to open burning. In the pathway: dispersion helps only a little; the honest answer is that the pathway cannot be cleaned, only shortened by reducing emissions. At the receptor: masks (a good one, worn properly), indoor air filtration, school closures on the worst days, and asthma plans for the vulnerable. A teacher's tools are mostly at the receptor

end: the AQI check, the morning timing of sports, the classroom with windows shut on a hazardous day.

Common misconception. "Clean air means air you cannot see," and its winter cousin, "smog is a winter problem only, and only in Lahore." Both halves of both claims fail. Lahore's smog is visible, but the most dangerous pollutant there, PM_{2.5}, is invisible particle by particle; a clear sky can still carry an unhealthy index. And the same pollutants burden Karachi's air and the summer afternoons' ozone, so the eye and the calendar are not measuring instruments either. The AQI is.

Activity: Read the index

Work in pairs. Your teacher puts one week of AQI readings for a Pakistani city on the board. For each day, name the band (good, moderate, unhealthy, hazardous), and write one sentence of school advice for that day. Then answer: which day would you cancel the sports day, and which would you move it to the morning? Allow 20 minutes.

Check your understanding

1. Name the six pollutants in this topic's table with their main sources.
(*Remember*)
2. Why is PM_{2.5} more dangerous than PM₁₀? Use the idea of where each can reach in the body. (*Understand*)
3. Explain why "the wind will clean it" is a weak control strategy, using the chain. (*Analyze*)
4. A headteacher asks what to do when the AQI reads "unhealthy". Give two actions and name the link of the chain each one acts on. (*Apply*)

Summary

Air pollution runs the same source-pathway-receptor chain as water pollution, with six pollutants that matter: particulates, nitrogen dioxide, sulfur dioxide, carbon monoxide, ozone and lead. The WHO estimates 4.2 million premature

deaths a year from outdoor air, and nearly everyone breathes air above its guidelines. Lahore's winter smog, trapped by inversions and fed by vehicles, kilns and burning, is the national case. Controls act at the source, weakly in the pathway, and at the receptor, where a teacher's AQI check and timing decisions live.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can name the six major pollutants and one effect of each.
- ☐ I can read an AQI band and say what a school should do in it.
- ☐ I can explain the inversion that traps Lahore's winter smog.
- ☐ I can name a control at each link of the air chain.

Try this at your practicum school

For one week, record the AQI for the school's city every morning at the same time, and one thing the school actually did in response, even if that was nothing. Bring back the seven readings and one sentence on whether the school's response matched the index. Do not name the school.

Summative task

Write 160 to 200 words on the following.

A parents' committee asks: "The smog is India's fault, so why should we change anything?" Using the chain, the pollutant table and the Lahore case, give an honest answer that names what is shared, what is ours, and what a school can still do.

Mini-rubric

Criterion	What a strong response does
Chain used (Analyze)	Separates sources that are shared from controls that still work
Evidence (Apply)	Uses the pollutant table and the study's findings, not opinion
School actions (Understand)	Names receptor-end actions a school can actually take
Clarity and register	Plain sentences for parents; within the word count

Further reading

- World Health Organization. (2024). *Ambient (outdoor) air pollution* (fact sheet): [https://www.who.int/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health)
- World Health Organization. (2021). *WHO global air quality guidelines*: <https://www.who.int/publications/i/item/9789240034228>
- Anwar, A., Younes, I., Afzal, M. Z., & Arshad, M. Z. (2026). Integrated remote sensing and socio-economic evaluation of winter smog and its impacts in Lahore district, Pakistan. *Environmental Systems Research*, 15(1). <https://doi.org/10.1186/s40068-026-00484-0>
- Holland, W. W. (Ed.). *Oxford Textbook of Public Health*. (guide-recommended book's closest real work, cited by reference only; flagged per D-2026-0001)

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